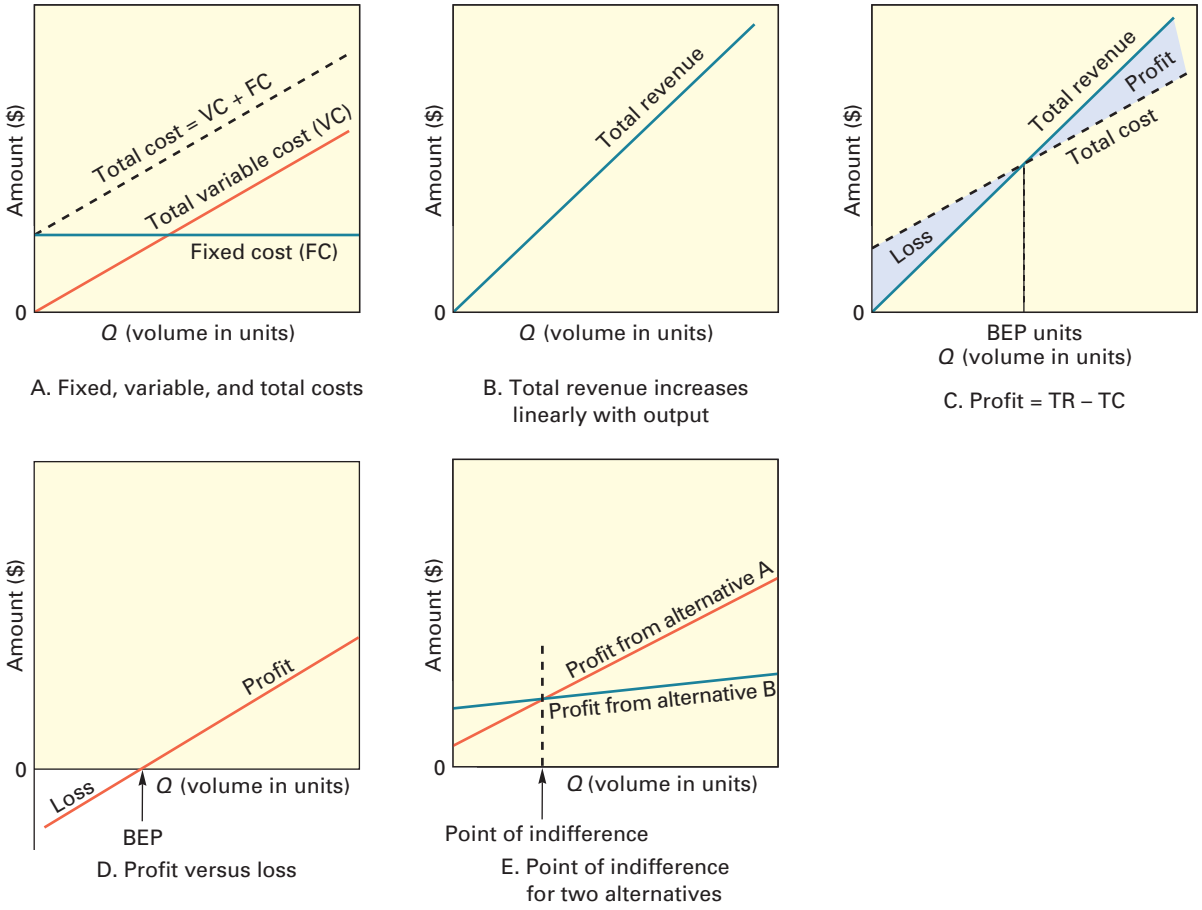
FIGURE 5.6
Cost–volume relationships



$$Q_{\text{BEP}} = \frac{\text{FC}}{R - v} \tag{5–9}$$

Different alternatives can be compared by plotting the profit lines for the alternatives, as shown in Figure 5.6E.

Figure 5.6E illustrates the concept of an **indifference point**: the quantity at which a decision maker would be indifferent between two competing alternatives. In this illustration, a quantity less than the point of indifference would favor choosing alternative B because its profit is higher in that range, while a quantity greater than the point of indifference would favor choosing alternative A.

Indifference point The quantity that would make two alternatives equivalent.

EXAMPLE 3
eXcel
mhhe.com/stevenson12e

The owner of Old-Fashioned Berry Pies, S. Simon, is contemplating adding a new line of pies, which will require leasing new equipment for a monthly payment of $6,000. Variable costs would be $2 per pie, and pies would retail for $7 each.

a. How many pies must be sold in order to break even?
b. What would the profit (loss) be if 1,000 pies are made and sold in a month?
c. How many pies must be sold to realize a profit of $4,000?
d. If 2,000 can be sold, and a profit target is $5,000, what price should be charged per pie?

PROBLEMS

1. Determine the utilization and the efficiency for each of these situations:
   a. A loan processing operation that processes an average of 7 loans per day. The operation has a design capacity of 10 loans per day and an effective capacity of 8 loans per day.
   b. A furnace repair team that services an average of four furnaces a day if the design capacity is six furnaces a day and the effective capacity is five furnaces a day.
   c. Would you say that systems that have higher efficiency ratios than other systems will always have higher utilization ratios than those other systems? Explain.
2. In a job shop, effective capacity is only 50 percent of design capacity, and actual output is 80 percent of effective output. What design capacity would be needed to achieve an actual output of eight jobs per week?
3. A producer of pottery is considering the addition of a new plant to absorb the backlog of demand that now exists. The primary location being considered will have fixed costs of $9,200 per month and variable costs of 70 cents per unit produced. Each item is sold to retailers at a price that averages 90 cents.
   a. What volume per month is required in order to break even?
   b. What profit would be realized on a monthly volume of 61,000 units? 87,000 units?
   c. What volume is needed to obtain a profit of $16,000 per month?
   d. What volume is needed to provide a revenue of $23,000 per month?
   e. Plot the total cost and total revenue lines.
4. A small firm intends to increase the capacity of a bottleneck operation by adding a new machine. Two alternatives, A and B, have been identified, and the associated costs and revenues have been estimated. Annual fixed costs would be $40,000 for A and $30,000 for B; variable costs per unit would be $10 for A and $11 for B; and revenue per unit would be $15.
   a. Determine each alternative's break-even point in units.
   b. At what volume of output would the two alternatives yield the same profit?
   c. If expected annual demand is 12,000 units, which alternative would yield the higher profit?
5. A producer of felt-tip pens has received a forecast of demand of 30,000 pens for the coming month from its marketing department. Fixed costs of $25,000 per month are allocated to the felt-tip operation, and variable costs are 37 cents per pen.
   a. Find the break-even quantity if pens sell for $1 each.
   b. At what price must pens be sold to obtain a monthly profit of $15,000, assuming that estimated demand materializes?
6. A real estate agent is considering changing her cell phone plan. There are three plans to choose from, all of which involve a monthly service charge of $20. Plan A has a cost of $.45 a minute for daytime calls and $.20 a minute for evening calls. Plan B has a charge of $.55 a minute for daytime calls and $.15 a minute for evening calls. Plan C has a flat rate of $80 with 200 minutes of calls allowed per month and a charge of $.40 per minute beyond that, day or evening.
   a. Determine the total charge under each plan for this case: 120 minutes of day calls and 40 minutes of evening calls in a month.
   b. Prepare a graph that shows total monthly cost for each plan versus daytime call minutes.
   c. If the agent will use the service for daytime calls, over what range of call minutes will each plan be optimal?
   d. Suppose that the agent expects both daytime and evening calls. At what point (i.e., percentage of call minutes for daytime calls) would she be indifferent between plans A and B?
7. A firm plans to begin production of a new small appliance. The manager must decide whether to purchase the motors for the appliance from a vendor at $7 each or to produce them in-house. Either of two processes could be used for in-house production; one would have an annual fixed cost of $160,000 and a variable cost of $5 per unit, and the other would have an annual fixed cost of $190,000 and a variable cost of $4 per unit. Determine the range of annual volume for which each of the alternatives would be best.
8. A manager is trying to decide whether to purchase a certain part or to have it produced internally. Internal production could use either of two processes. One would entail a variable cost of $17 per unit and an annual fixed cost of $200,000; the other would entail a variable cost of $14 per unit and an annual fixed cost of $240,000. Three vendors are willing to provide the part. Vendor A has a price of $20 per unit for any volume up to 30,000 units. Vendor B has a price of $22 per unit for demand of 1,000 units or less, and $18 per unit for larger quantities. Vendor C offers a price of $21 per unit for the first 1,000 units, and $19 per unit for additional units.

DISCUSSION AND REVIEW QUESTIONS

1. What is the chief role of the operations manager?
2. List the steps in the decision-making process.
3. Explain the term *bounded rationality.*
4. Explain the term *suboptimization.*
5. What are some of the reasons for poor decisions?
6. What information is contained in a payoff table?
7. What is sensitivity analysis, and how can it be useful to a decision maker?
8. Contrast maximax and maximin decision strategies. Under what circumstances is each appropriate?
9. Under what circumstances is expected monetary value appropriate as a decision criterion? When isn't it appropriate?
10. Explain or define each of these terms:
    a. Laplace criterion.
    b. Minimax regret.
    c. Expected value.
    d. Expected value of perfect information.
11. What information does a decision maker need in order to perform an expected-value analysis of a problem? What options are available to the decision maker if the probabilities of the states of nature are unknown? Can you think of a way you might use sensitivity analysis in such a case?
12. Suppose a manager is using maximum EMV as a basis for making a capacity decision and, in the process, obtains a result in which there is a virtual tie between two of the seven alternatives. How is the manager to make a decision?
13. Identify three potential unethical actions or inactions related to decision analysis and the ethical principle each violates (see Chapter 1).

PROBLEMS

1. A small building contractor has recently experienced two successive years in which work opportunities exceeded the firm's capacity. The contractor must now make a decision on capacity for next year. Estimated profits under each of the two possible states of nature are as shown in the table below. Which alternative should be selected if the decision criterion is
    a. Maximax?
    b. Maximin?
    c. Laplace?
    d. Minimax regret?

	NEXT YEAR'S DEMAND	
Alternative	**Low**	**High**
Do nothing	$50*	$60
Expand	20	80
Subcontract	40	70

*Profit in $ thousands.

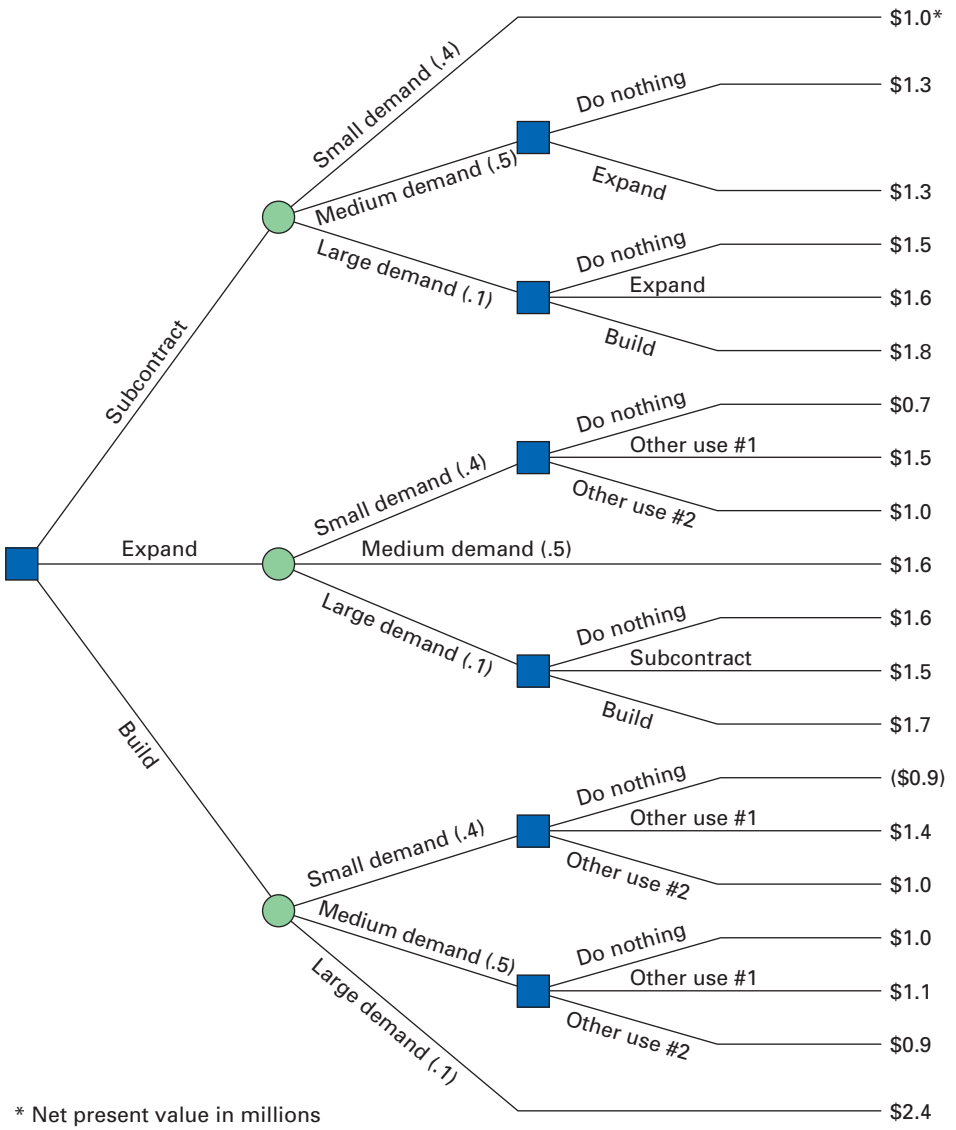
2. Refer to Problem 1. Suppose after a certain amount of discussion, the contractor is able to subjectively assess the probabilities of low and high demand: $P(\text{low}) = .3$ and $P(\text{high}) = .7$.
    a. Determine the expected profit of each alternative. Which alternative is best? Why?
    b. Analyze the problem using a decision tree. Show the expected profit of each alternative on the tree.
    c. Compute the expected value of perfect information. How could the contractor use this knowledge? 260
3. Refer to Problems 1 and 2. Construct a graph that will enable you to perform sensitivity analysis on the problem. Over what range of $P(\text{high})$ would the alternative of doing nothing be best? Expand? Subcontract?
4. A firm that plans to expand its product line must decide whether to build a small or a large facility to produce the new products. If it builds a small facility and demand is low, the net present value after deducting for building costs will be $400,000. If demand is high, the firm can either maintain the small facility or expand it. Expansion would have a net present value of $450,000, and maintaining the small facility would have a net present value of $50,000.

If a large facility is built and demand is high, the estimated net present value is $800,000. If demand turns out to be low, the net present value will be −$10,000.

The probability that demand will be high is estimated to be .60, and the probability of low demand is estimated to be .40.

a. Analyze using a tree diagram.
b. Compute the EVPI. How could this information be used?
c. Determine the range over which each alternative would be best in terms of the value of P (demand low).

5. Determine the course of action that has the highest expected payoff for this decision tree.



* Net present value in millions

6. The lease of Theme Park, Inc., is about to expire. Management must decide whether to renew the lease for another 10 years or to relocate near the site of a proposed motel. The town planning board is currently debating the merits of granting approval to the motel. A consultant has estimated the net present value of Theme Park's two alternatives under each state of nature as shown on the following page.

What course of action would you recommend using?

a. Maximax.
b. Maximin.
c. Laplace.
d. Minimax regret.

Options	Motel Approved	Motel Rejected
Renew	$ 500,000	$4,000,00
Relocate	5,000,000	100,00

7. Refer to Problem 6. Suppose that the management of Theme Park, Inc., has decided that there is a .35 probability that the motel's application will be approved.

a. If management uses maximum expected monetary value as the decision criterion, which alternative should it choose?
b. Represent this problem in the form of a decision tree.
c. If management has been offered the option of a temporary lease while the town planning board considers the motel's application, would you advise management to sign the lease? The lease will cost $24,000.

8. Construct a graph that can be used for sensitivity analysis for the preceding problem.

a. How sensitive is the solution to the problem in terms of the probability estimate of .35?
b. Suppose that, after consulting with a member of the town planning board, management decides that an estimate of approval is approximately .45. How sensitive is the solution to this revised estimate? Explain.
c. Suppose the management is confident of all the estimated payoffs except for $4 million. If the probability of approval is .35, for what range of payoff for renew/reject will the alternative selected using the maximum expected value remain the same?

9. A firm must decide whether to construct a small, medium, or large stamping plant. A consultant's report indicates a .20 probability that demand will be low and an .80 probability that demand will be high.

If the firm builds a small facility and demand turns out to be low, the net present value will be $42 million. If demand turns out to be high, the firm can either subcontract and realize the net present value of $42 million or expand greatly for a net present value of $48 million.

The firm could build a medium-size facility as a hedge: If demand turns out to be low, its net present value is estimated at $22 million; if demand turns out to be high, the firm could do nothing and realize a net present value of $46 million, or it could expand and realize a net present value of $50 million.

If the firm builds a large facility and demand is low, the net present value will be −$20 million, whereas high demand will result in a net present value of $72 million.

a. Analyze this problem using a decision tree.
b. What is the maximin alternative?
c. Compute the EVPI and interpret it.
d. Perform sensitivity analysis on *P*(high).

10. A manager must decide how many machines of a certain type to buy. The machines will be used to manufacture a new gear for which there is increased demand. The manager has narrowed the decision to two alternatives: buy one machine or buy two. If only one machine is purchased and demand is more than it can handle, a second machine can be purchased at a later time. However, the cost per machine would be lower if the two machines were purchased at the same time.

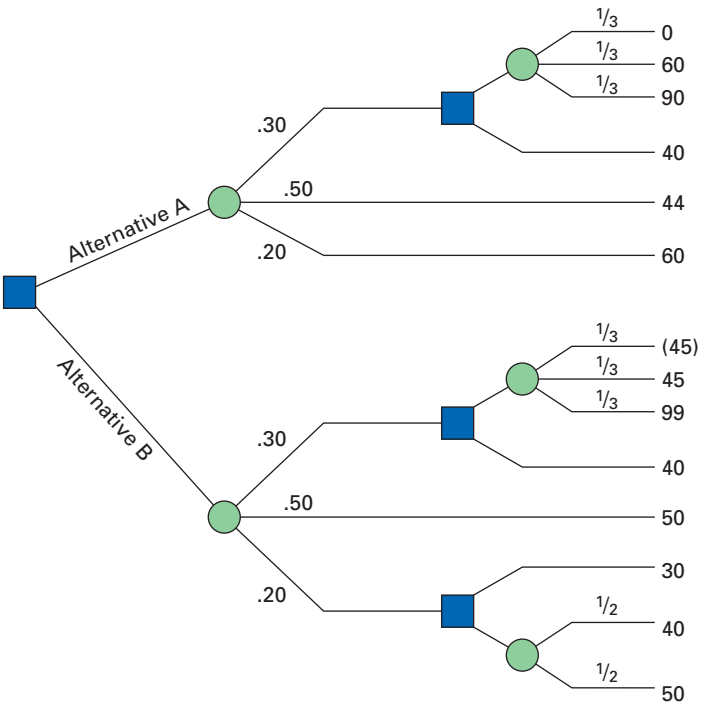
The estimated probability of low demand is .30, and the estimated probability of high demand is .70.

The net present value associated with the purchase of two machines initially is $75,000 if demand is low and $130,000 if demand is high.

The net present value for one machine and low demand is $90,000. If demand is high, there are three options. One option is to do nothing, which would have a net present value of $90,000. A second option is to subcontract; that would have a net present value of $110,000. The third option is to purchase a second machine. This option would have a net present value of $100,000.

How many machines should the manager purchase initially? Use a decision tree to analyze this problem.

11. Determine the course of action that has the highest EMV for the accompanying tree diagram.



12. A logistics provider plans to have a new warehouse built to handle increasing demands for its services. Although the company is unsure of how much demand there will be, it must decide now on the size (large or small) of the warehouse. Preliminary estimates are that if a small warehouse is built and demand is low, the monthly income will be $700,000. If demand is high, it will have to either expand the facility or lease additional space. Leasing will result in a monthly income of $100,000 while expanding will result in a monthly income of $500,000.

If a large warehouse is built and demand is low, monthly income will only be $40,000, while if demand is high, monthly income will be $2 million.

a. Construct a tree diagram for this decision.
b. Using your tree diagram, identify the choice that would be made using each of the four approaches for decision making under uncertainty.

13. The director of social services of a county has learned that the state has mandated additional information requirements. This will place an additional burden on the agency. The director has identified three acceptable alternatives to handle the increased workload. One alternative is to reassign present staff members, the second is to hire and train two new workers, and the third is to redesign current practice so that workers can readily collect the information with little additional effort. An unknown factor is the caseload for the coming year when the new data will be collected on a trial basis. The estimated costs for various options and caseloads are shown in the following table:

	CASELOAD		
	Moderate	**High**	**Very High**
Reassign staff	$50*	60	85
New staff	60	60	60
Redesign collection	40	50	90

*Cost in $ thousands.

Assuming that past experience has shown the probabilities of various caseloads to be unreliable, what decision would be appropriate using each of the following criteria?

a. Maximin.
b. Maximax.
c. Minimax regret.
d. Laplace.

DISCUSSION AND REVIEW QUESTIONS

1. A project manager may need two skill sets—those of a manager and those of a leader. Explain.
2. Explain the term *project champion* and list some ways to keep a champion involved with the project.
3. List the steps in risk management.
4. Give some examples of ethical issues that may arise on projects. What can a project manager do to minimize such issues?
5. What are the key advantages of using project management software?
6. What is a work breakdown structure, and how is it useful for project planning?
7. Identify the term being described for each of the following:
   a. A sequence of activities in a project.
   b. The longest time sequence of activities in a project.
   c. Used when two activities have the same starting and finishing points.
   d. The difference in time length of any path and the critical path.
   e. The statistical distribution used to describe variability of an activity time.
   f. The statistical distribution used to describe path variability.
   g. Shortening an activity by allocating additional resources.
8. List the main advantages of PERT. List the main limitations.
9. Why might a probabilistic estimate of a project's completion time based solely on the variance of the *critical path* be misleading? Under what circumstances would it be acceptable?
10. Define each of these terms, and indicate how each is determined.
    a. Expected activity time.
    b. Variance of an activity time.
    c. Standard deviation of a path's time.
11. Why might a person wish to be involved with a critical path activity? What are some of the reasons one might have for not wanting this association?
12. What are some of the potential benefits of working on a special project in one's firm? What are some of the risks?
13. What are some aspects of the project manager's job that make it more demanding than the job of a manager working in a more routine organizational framework?
14. What is the main benefit of a project organization over more traditional forms of operations management for project work?

TAKING STOCK

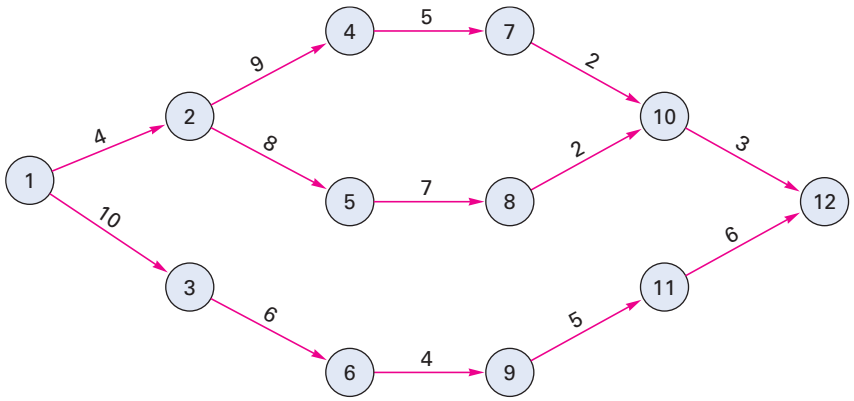
1. What trade-offs are associated with time and cost estimates for a proposed project?
2. Who needs to be involved in assessing the cost of a project?
3. Name and explain briefly two ways that technology has had an impact on project management.

CRITICAL THINKING EXERCISES

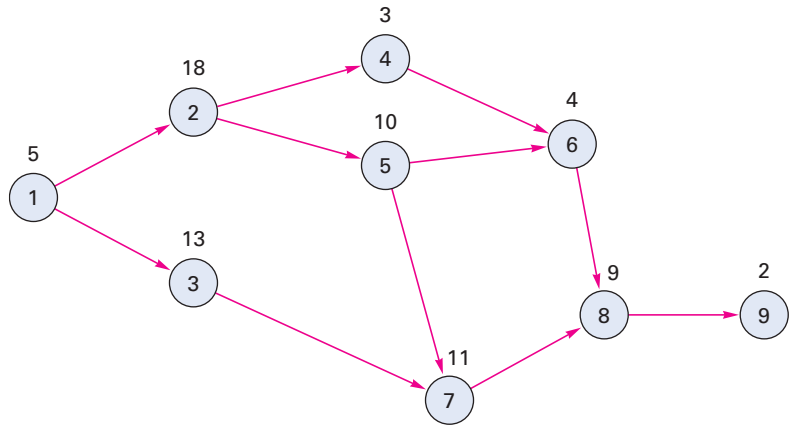
1. Project management techniques have been used successfully for a wide variety of efforts, including the many NASA space missions, huge construction projects, implementation of major systems such as ERP, production of movies, development of new products and services, theatrical productions, and much more. Why not use them for managing the operations function of any business?
2. Give three examples of unethical conduct involving projects and the ethical principle each one violates.

PROBLEMS

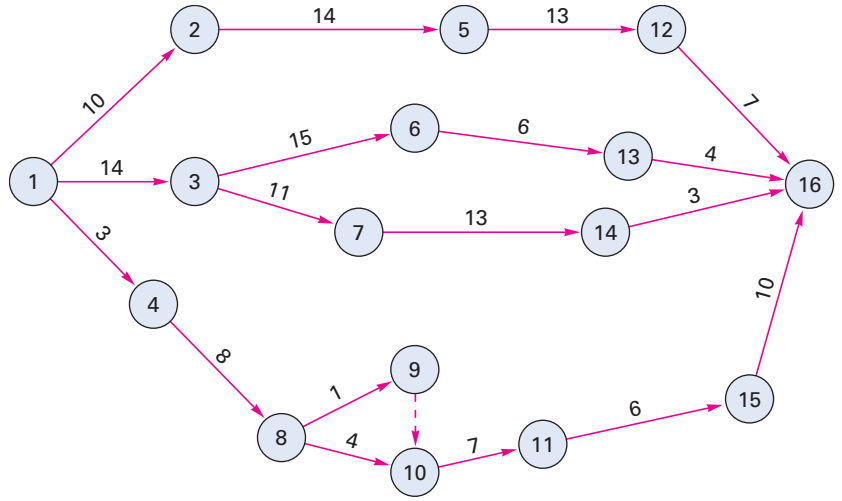
1. For each of the following network diagrams, determine both the critical path and the expected project duration. The numbers on the arrows represent expected activity times.
   a. AOA diagram



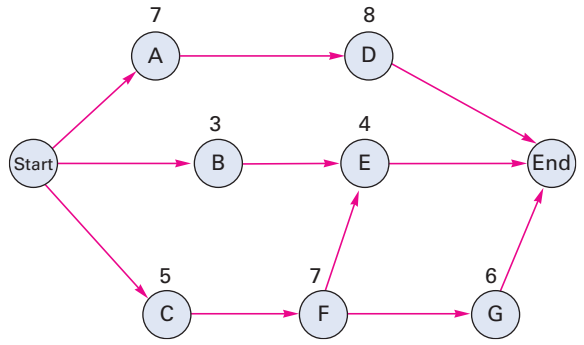
b. AON diagram



c. AOA diagram



d. AON diagram



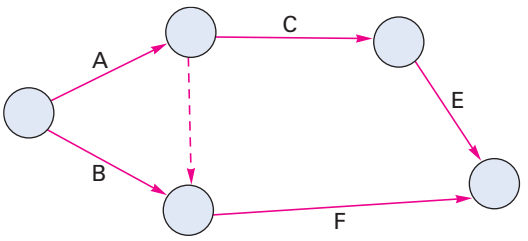
2. Chris received new word processing software for her birthday. She also received a check, with which she intends to purchase a new computer. Chris's college instructor assigned a paper due next week. Chris decided that she will prepare the paper on the new computer. She made a list of the activities she will need to do and their estimated times.
   a. Arrange the activities into two logical sequences.
   b. (1) Construct an AOA network diagram.
      (2) Construct an AON diagram.
   c. Determine the critical path and the expected duration time.

d. What are some possible reasons for the project to take longer than the expected duration?

Estimated Time (hrs.)	Activity (abbreviation)
0.8	Install software (Inst)
0.4	Outline the paper (Out)
0.2	Submit paper to instructor (Sub)
0.6	Choose a topic (Ch)
0.5	Use grammar-checking routine and make corrections (Ck)
3.0	Write the paper using the word-processing software (Write)
2.0	Shop for a new computer (Sh)
1.0	Select and purchase computer (Sel)
2.0	Do library research on chosen topic (Lib)

3. Prepare a Gantt chart for each of the following in the style of the chart shown on p. 739.
   a. The bank location problem (see Figure 17.4, p. 741). *Hint:* Use the early start (ES) times given in Table 17.3 on p. 751.
   b. Solved Problem number 2 on pp. 766–768.

4. a. Develop a list of activities and their immediate predecessors similar to the lists in this problem for this diagram:



   b. Construct an activity-on-arrow precedence diagram for each of the following cases. Note that each case requires the use of a dummy activity.
   c. Construct an AON diagram for each case.

(1) Activity	Immediate Predecessor	(2) Activity	Immediate Predecessor
A	—	J	—
B	—	K	—
C	—	L	J
D	A	M	L
E	B	N	J
F	B	P	N
G	C	Q	—
H	F	R	K
I	F, G	S	Q
K	D, E	V	R, S, T
End	H, I, K	T	Q
		W	T
		End	M, P, V, W

5. For each of the problems listed, determine the following quantities for each activity: the earliest start time, latest start time, earliest finish time, latest finish time, and slack time. List the critical activities, and determine the expected duration of the project.
   a. Problem 1*a*.
   b. Problem 1*b*.

4. The owner of Eat Now Restaurant implemented an expanded menu early last year. The menu was a success, drawing many more customers, who seemed to like the increased variety of menu choices over that of the previous menu. But good news soon became bad news as long waiting lines began to deter customers, and business dropped off. Because of space and other limitations, there didn't seem to be any viable options to consider. Then a customer mentioned a technique called mass customization that was being used in the company he worked for. He said it really streamlined processing, and maybe it could work for the restaurant.

   Describe how that approach might work at the restaurant and why that could be expected to reduce waiting times. What costs would be involved in transitioning to such a system? What other approaches could be used to reduce waiting times?

5. Describe two examples of unethical behavior related to waiting line management, and state which ethical principles they violate.

PROBLEMS

1. a. $\lambda = 3$ customers/hour
   $\mu = 5$ customers/hour
   $M = 1$
   1) What is the system utilization?
   2) What is the average number of customers waiting for service?
   3) What is the average time customers wait in line for service?

   b. Repair calls are handled by one repairman at a photocopy shop. Repair time, including travel time, is exponentially distributed, with a mean of two hours per call. Requests for copier repairs come in at a mean rate of three per eight-hour day (assume Poisson). Determine:
   1) The average number of customers awaiting repairs.
   2) System utilization.
   3) The amount of time during an eight-hour day that the repairman is not out on a call.
   4) The probability of two or more customers in the system.

   c. An average of 18 customers arrive at a service center each hour. There are two servers on duty, and each server can process 12 customers per hour.
   1) What is the system utilization?
   2) What is the average number of customers in the system (waiting plus being served)?
   3) What is the average time customers wait in line for service?
   4) What is the average waiting time for customers who actually have to wait?

2. A vending machine dispenses hot chocolate or coffee. Service time is 30 seconds per cup and is constant. Customers arrive at a mean rate of 80 per hour, and this rate is Poisson-distributed. Determine:
   a. The average number of customers waiting in line.
   b. The average time customers spend in the system.
   c. The average number in the system.

3. Many of a bank's customers use its automatic teller machine to transact business after normal banking hours. During the early evening hours in the summer months, customers arrive at a certain location at the rate of one every other minute. This can be modeled using a Poisson distribution. Each customer spends an average of 90 seconds completing his or her transactions. Transaction time is exponentially distributed. Determine:
   a. The average time customers spend at the machine, including waiting in line and completing transactions.
   b. The probability that a customer will not have to wait upon arriving at the automatic teller machine.
   c. The average number waiting to use the machine.

4. The following information pertains to telephone calls to a motel switchboard on a typical Tuesday.

Period	Incoming Rate (calls per minute)	Service Rate (calls per minute per operator)	Number of Operators
Morning	1.8	1.5	2
Afternoon	2.2	1.0	3
Evening	1.4	0.7	3